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SECTION: BSIS02	

- Provide a clear definition and overview of the following:
- Data Structures and algorithm
 - Data Structures
 - Data structure refers to an approach for arranging and storing data such that a program can access, manipulate, and organize the data effectively. Some examples of data structures include arrays, linked lists, stacks, queues, trees, graphs, and hash tables. Data structures vary based on the purpose for which they are used and their efficiency.
 - Algorithms
 - Algorithm refers to the process through which the solution to a particular problem can be attained. In programming, an algorithm is what dictates the steps that should be taken by the computer in order to get the desired result. Algorithm analysis typically involves the amount of time and space required by the algorithm.
- Linear Data Structure
 - Linear data structure refers to the data structure wherein the elements are stored in a linear sequence manner. Normally, each element has predecessors and/or successors, with the exception of the first and last element.
 - Examples of linear data structures are as follows:
 - Array
 - Linked list
 - Stack

- Queue
- One-Dimensional Array
 - An array in one dimension is a linear data structure consisting of elements of the same type stored in a single list. Each element of the array can be accessed by its index. The index of the first element in a Java array is always 0.
- The Big O Notation
 - Big-O Notation is a mathematical notation used for representing the growth rate of resources used by an algorithm with the respect to increase in input size, especially in terms of running time or memory space. It emphasizes the growth rate of the algorithm but not the exact running time of the computer.
- Basic Searching Algorithm
 - A searching algorithm is any algorithm that is designed to search for a specific value, known as the target, within some data set.
 - Linear Search
 - Linear Search or Sequential Search examines each element one after another starting from the first element until either the target element is located or all elements have been examined.
 - ex. [15, 28, 34, 42, 57] ↑ Check 15 → Not found [15, 28, 34, 42, 57] ↑ Check 28 → Not found [15, 28, 34, 42, 57] ↑ Check 34 → Not found [15, 28, 34, 42, 57] ↑ Check 42 → FOUND!
 - Binary Search
 - Binary search is an algorithm used for searching that continually splits the search space in half. This differs from the linear search algorithm, which does not require sorting.

- ex. [15, 28, 34, 42, 57] $\uparrow$ Middle = 34 $42 > 34$ Therefore, ignore the left half. [42, 57] $\uparrow$ Middle = 42 $42 = 42$
FOUND!

References:

<https://docs.oracle.com/javase/tutorial/java/nutsandbolts/arrays.html>

https://opensa-server.cs.vt.edu/ODSA/Books/pubbook/odsa-all/fall-2019/Public_Instance/html/BinarySearch.html

<https://opensa.cs.vt.edu/ODSA/Books/SWDesignAndDataStructs/html/ContentBinarySearch.html>